

Trends in University Department Preferences in Turkey (2018–2025)

Muhammed Reha Özdemir and Yeşim Kaya

1. Project Overview and Scope

Abstract: This study analyzes university department preference trends in Turkey between 2018 and 2025 using YÖK Atlas data. Seven major field categories are examined through admission scores and program counts. The findings show that medicine and dentistry/pharmacy remain the most competitive fields throughout the period, computer engineering leads among engineering sub-fields with a strong upward trend toward 2025, and state universities maintain a slight score advantage over private ones. A simple linear model is also applied to project future score trends for computer engineering. The results offer a data-driven perspective on how student preferences and higher education supply have evolved over time in Turkey.

This project examines how preferences for university departments in Turkey have shifted over the years between 2018 and 2025. Using data from YÖK Atlas — the official platform maintained by the Council of Higher Education — we track admission scores and program counts across seven major field categories: engineering, medicine, law, psychology, teaching, architecture, and dentistry/pharmacy. The central question we try to answer is straightforward: which fields are gaining or losing popularity among students, and what might be driving these changes?

Our objectives for this study are as follows:

- To identify demand trends at the category level across the 2018–2025 period
- To compare engineering sub-fields, particularly computer engineering versus civil and industrial engineering
- To investigate the score gap between state and private university programs
- To explore how programs are distributed across cities in Turkey
- To build a simple trend model and estimate where admission scores might be heading

2. Data

2.1 Data Sources

Three data sources were combined to cover the full 2018–2025 period. The first is the `thstats` R package developed by Mustafa Cavus and Olgun Aydin, which provides structured YÖK Atlas data for the years 2018–2020. While this package makes early-period data conveniently accessible, its coverage ends at 2020. To fill the 2021 gap, we used a third-party dataset shared on GitHub by MorphaxTheDeveloper, which was compiled from YÖK Atlas records for that year. For the more recent years between 2022 and 2025, we pulled data directly from the YÖK Atlas JSON API using the `httr2` package in R. Among the three sources, the API data is the most reliable since it comes straight from the official system, while the third-party 2021 dataset required extra verification during preprocessing.

References: YÖK Atlas (<https://yokatlas.yok.gov.tr/>), thestats package (<https://github.com/analyticsresearchlab/thestats>), MorphaxTheDeveloper dataset (<https://github.com/MorphaxTheDeveloper/yokatlas-dataset-2025>)

2.2 Preprocessing

Since the three sources use slightly different column names and encoding conventions, we merged them into a single table and standardized department names by matching Turkish and English variants. University type labels (`unitur`) were also unified, and a `bolum_grubu` column was added to group programs into broader sub-field categories for easier comparison.

```
# Read the combined dataset
df <- fread("yokatlas_FINAL_CLEAN.csv", encoding = "UTF-8")

# Convert to tibble for tidyverse compatibility
df <- as_tibble(df)

# Quick structure check
cat("Dataset dimensions:", nrow(df), "rows x", ncol(df), "columns\n")
```

Dataset dimensions: 75145 rows x 11 columns

```
cat("Years covered:", paste(sort(unique(df$year)), collapse = ", "), "\n")
```

Years covered: 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025

```
cat("Categories:", paste(sort(unique(df$kategori)), collapse = ", "), "\n")
```

Categories: Diger, Dis_Hekimligi_Eczacilik, Hukuk, Mimarlik, Muhendislik, Ogretmenlik, Psikolo

```
# Save as .RData for reproducibility
save(df, file = "data.RData")
```

The processed dataset is stored in `.RData` format and can be accessed [here](#).

2.3 Data Overview

The dataset contains the following key columns:

Column	Description
year	Academic year of admission
kategori	Field category (e.g., Muhendislik, Tip)
bolum_grubu	Engineering sub-field grouping
il	City where the university is located
unitur	University type: DEVLET (state) or VAKIF (private)
puan	Minimum admission score for the program
sira	Admission rank corresponding to the minimum score
kontenjan	Total quota (capacity) of the program

Column	Description
yerlesen	Number of students actually enrolled

```
# Programs per year
df |>
  group_by(year) |>
  summarise(n_programs = n(),
            avg_puan = round(mean(puan, na.rm = TRUE), 1),
            avg_sira = round(mean(sira, na.rm = TRUE), 0)) |>
  print()
```

```
# A tibble: 8 x 4
  year n_programs avg_puan avg_sira
<int>   <int>     <dbl>   <dbl>
1  2018     5935     323.    198128
2  2019     6764     324.    239592
3  2020     9571     318.    312161
4  2021    18357     175    329515
5  2022     8354     284    359386
6  2023     8927     272.    382464
7  2024     9640     243.    391498
8  2025     7597     334.    336911
```

```
# Programs per category
df |>
  filter(kategori != "Diger") |>
  group_by(kategori) |>
  summarise(n_programs = n(),
            avg_puan = round(mean(puan, na.rm = TRUE), 1)) |>
  arrange(desc(avg_puan)) |>
  print()
```

```
# A tibble: 7 x 3
  kategori          n_programs avg_puan
<chr>             <int>     <dbl>
1 Tip              1323      433.
2 Dis_Hekimligi_Eczacilik 1673      396.
3 Hukuk            1178      350.
4 Ogretmenlik      4204      323.
5 Psikoloji        2801      306.
6 Muhendislik     13495      291.
7 Mimarlik         3106      266.
```

The dataset covers eight academic years and contains records from all seven field categories. Medicine has the highest average admission score at 433, followed by dentistry/pharmacy at 396, while architecture sits at the bottom with 266. Engineering, despite having the largest number of programs by far — over 13,000 records across all years — averages around 291, which reflects the wide range of programs it includes. The 2021 row shows an unusually high program count of 18,357, which is likely an artifact of how the third-party data source for that year was structured rather than a real spike in offerings.

3. Analysis

3.1 Exploratory Data Analysis

```
df |>
  filter(kategori != "Diğer", year == 2024) |>
  ggplot(aes(x = kategori, y = puan, fill = kategori)) +
  geom_boxplot(show.legend = FALSE) +
  labs(title = "Score Distribution by Category (2024)",
       x = NULL, y = "Minimum Admission Score") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 25, hjust = 1))
```

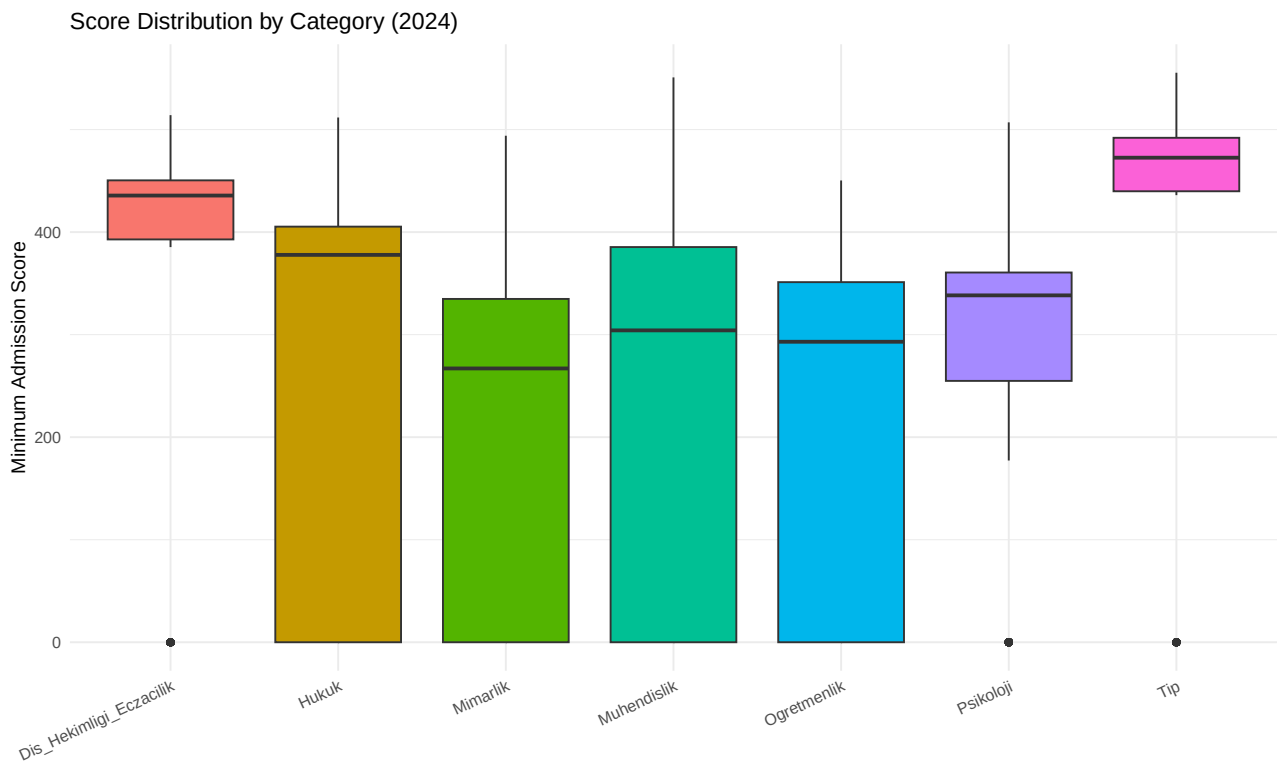


Figure 1: Minimum admission score distribution across field categories in 2024.

As shown in Figure 1, the 2024 score distributions show considerable variation both across and within categories. Medicine and dentistry/pharmacy have the highest and most compact distributions, meaning most programs in these fields require similarly high scores. Law and engineering show wider spreads, with engineering in particular covering a very broad range — from around 0 to well above 400 — which points to the coexistence of very selective programs and much lower-threshold ones within the same category. Psychology’s interquartile range sits mostly between 250 and 375, with a few outliers near zero that are likely low-quota private programs.

```
df |>
  filter(year == 2024, !is.na(puan), puan > 0) |>
  ggplot(aes(x = puan)) +
  geom_histogram(bins = 40, fill = "#2E86AB", color = "white") +
  labs(title = "Distribution of Minimum Admission Scores (2024)",
```

```
x = "Minimum Admission Score", y = "Count") +
theme_minimal()
```

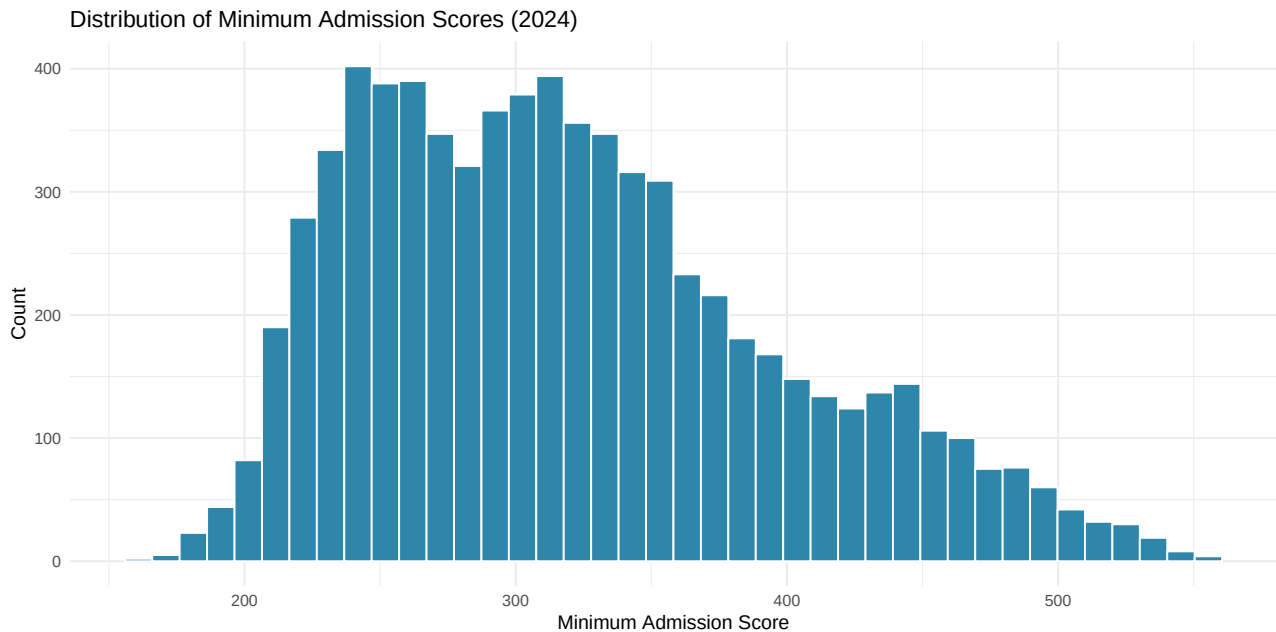


Figure 2: Histogram of minimum admission scores across all programs in 2024.

As shown in Figure 2, the overall distribution of admission scores in 2024 is roughly bell-shaped, with the highest concentration of programs falling between 250 and 350 points. A secondary cluster near the lower end of the scale represents programs that did not fully fill their quotas or have very low demand.

```
df |>
  summarise(
    kontenjan_na = sum(is.na(kontenjan)),
    yerlesen_na  = sum(is.na(yerlesen)),
    puan_na      = sum(is.na(puan)),
    sira_na      = sum(is.na(sira))
  ) |>
  print()
```

```
# A tibble: 1 x 4
  kontenjan_na yerlesen_na puan_na sira_na
      <int>      <int>   <int>   <int>
1      28148        4452     0      5357
```

Missing values are present mainly in the `yerlesen` (enrolled students) and `sira` (rank) columns, which is expected since not all programs fill their quotas every year and rank data is not always published for very low-scoring entries. The `puan` column, which is the primary variable used throughout the analysis, has a relatively small number of missing entries and these were excluded from calculations using `na.rm = TRUE`.

3.2 Category Trend Analysis

```
# Average minimum score by category over years
category_trend <- df |>
  filter(kategori != "Diger") |>
  group_by(year, kategori) |>
  summarise(avg_puan = mean(puan, na.rm = TRUE), .groups = "drop")

ggplot(category_trend, aes(x = year, y = avg_puan, color = kategori)) +
  geom_line(linewidth = 1.1) +
  geom_point(size = 2) +
  scale_x_continuous(breaks = 2018:2025) +
  labs(title = "Average Minimum Admission Score by Category (2018-2025)",
       x = "Year", y = "Average Min Score", color = "Category") +
  theme_minimal()
```

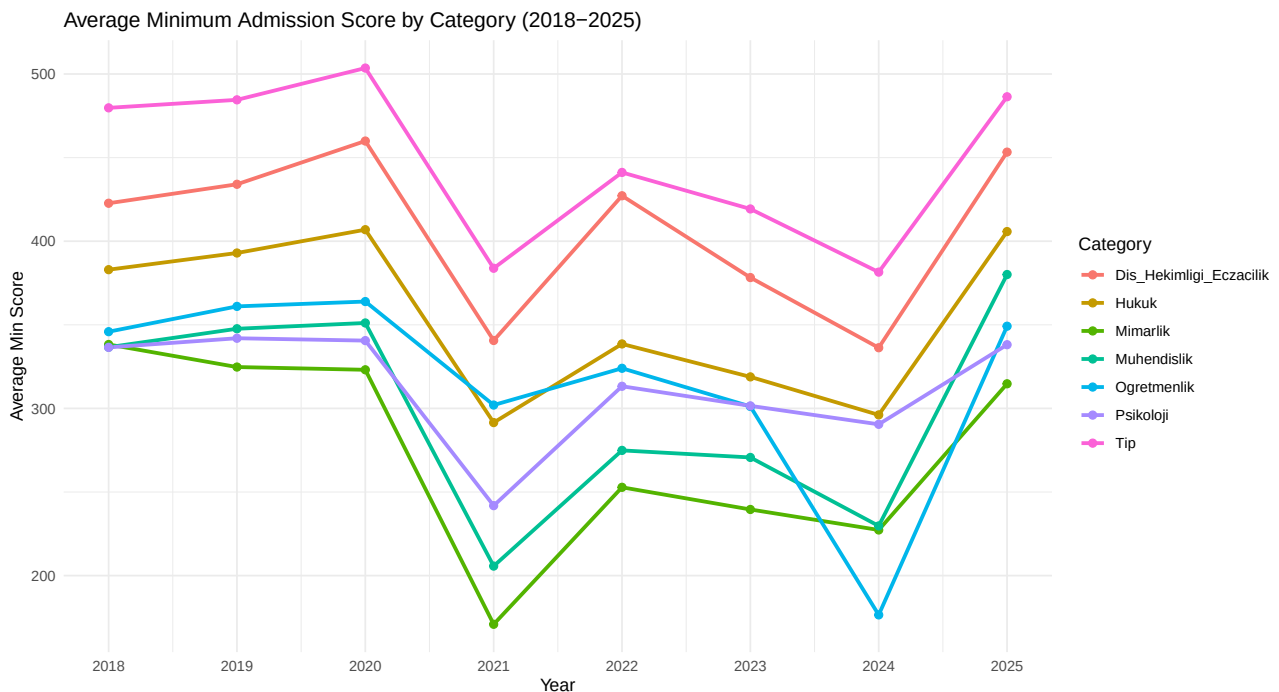


Figure 3: Average minimum admission score by field category from 2018 to 2025.

As shown in Figure 3, the most striking feature of this chart is the sharp drop across all categories in 2021, followed by a partial recovery in 2022. This pattern is consistent with a data source change for 2021 rather than a real-year effect, and should be interpreted with caution. Setting that aside, medicine and dentistry/pharmacy have maintained the highest scores throughout the period and both show an upward push toward 2025. Law recovered strongly after 2022 and is trending upward. Engineering and architecture have remained relatively flat, while psychology and teaching stay in the lower range with little clear directional movement.

```
# Program count trend
program_count <- df |>
  filter(kategori != "Diger") |>
  group_by(year, kategori) |>
  summarise(n = n(), .groups = "drop")
```

```
ggplot(program_count, aes(x = year, y = n, color = kategori)) +
  geom_line(linewidth = 1.1) +
  geom_point(size = 2) +
  scale_x_continuous(breaks = 2018:2025) +
  labs(title = "Number of Programs by Category (2018-2025)",
       x = "Year", y = "Number of Programs", color = "Category") +
  theme_minimal()
```

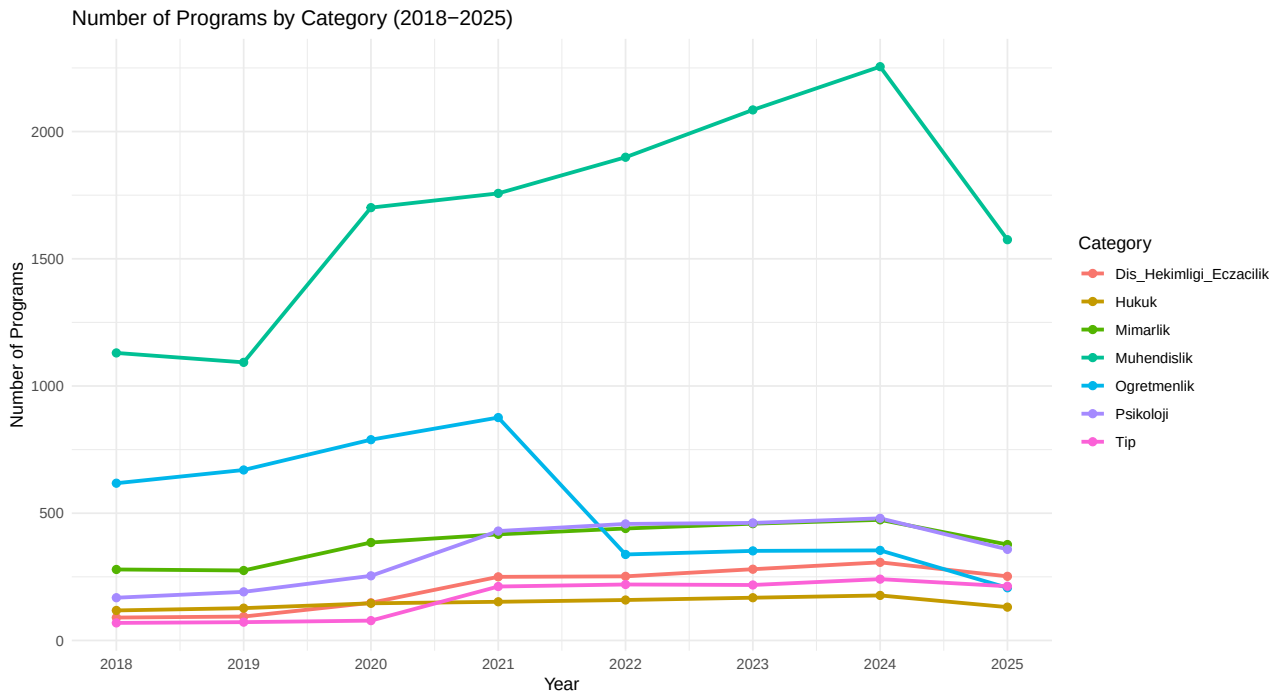


Figure 4: Number of programs per field category from 2018 to 2025.

As shown in Figure 4, engineering dominates the program count chart throughout the entire period, reaching a peak of around 2,300 programs in 2024 before dropping in 2025. Teaching is the second largest category, though it shows a sharp decline after 2021 — possibly due to quota reductions by the government. Psychology has grown modestly but steadily, going from near the bottom of the chart in 2018 to clearly separating itself from the smaller categories by 2024. Dentistry/pharmacy and medicine remain flat and low in count, which is consistent with the controlled and limited capacity of those fields.

3.3 Engineering Sub-field Analysis

```
# Top 6 engineering sub-fields by program count
top_eng <- df |>
  filter(kategori == "Muhendislik",
         bolum_grubu != "Diger Muhendislik") |>
  group_by(bolum_grubu) |>
  summarise(n = n()) |>
  arrange(desc(n)) |>
  head(6) |>
  pull(bolum_grubu)
```

```
# Score trend for top sub-fields
eng_trend <- df |>
  filter(bolum_grubu %in% top_eng) |>
  group_by(year, bolum_grubu) |>
  summarise(avg_puan = mean(puan, na.rm = TRUE), .groups = "drop")

ggplot(eng_trend, aes(x = year, y = avg_puan, color = bolum_grubu)) +
  geom_line(linewidth = 1.1) +
  geom_point(size = 2) +
  scale_x_continuous(breaks = 2018:2025) +
  labs(title = "Top 6 Engineering Sub-fields: Admission Score Trend",
       x = "Year", y = "Average Min Score", color = "Sub-field") +
  theme_minimal()
```

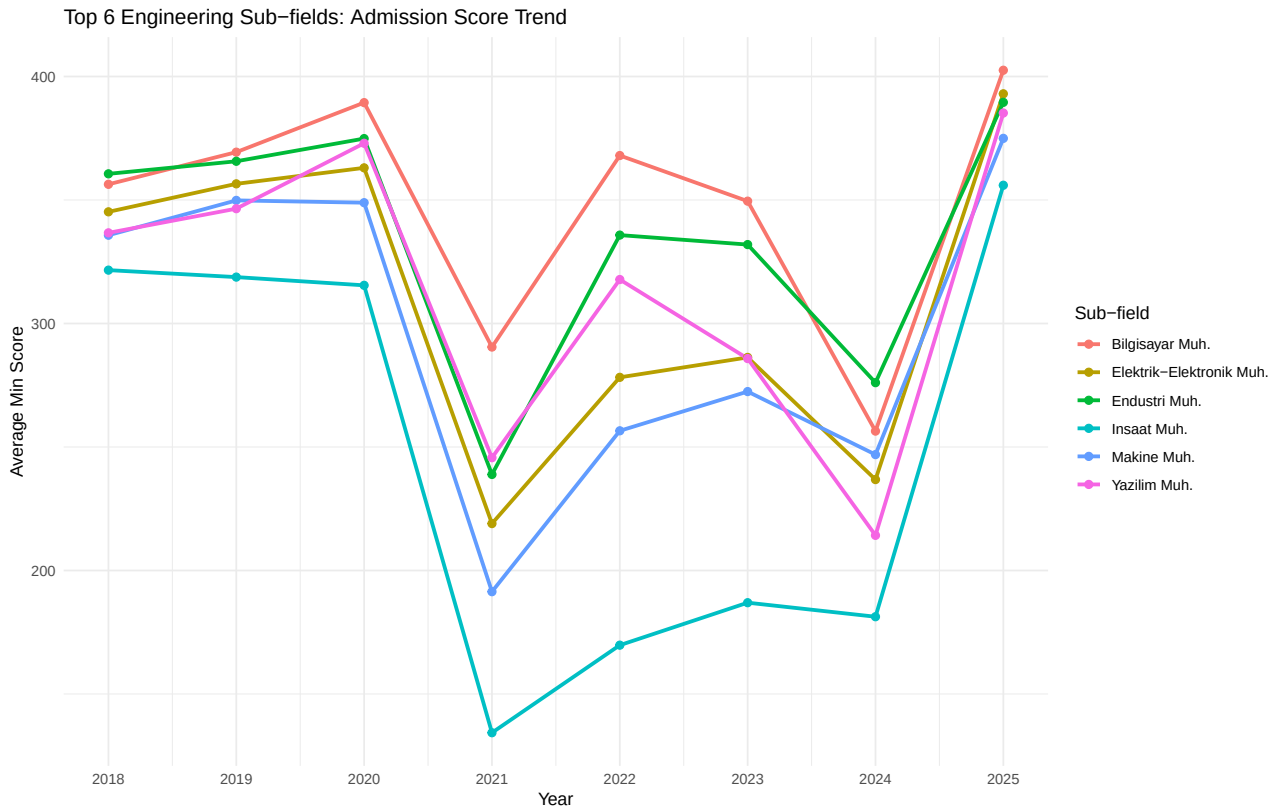


Figure 5: Average minimum admission score trend for the top 6 engineering sub-fields.

As shown in Figure 5, among the six sub-fields, computer engineering has held the highest average scores for most of the period and shows a strong recovery toward 2025, reaching around 400. Software engineering follows a similar upward trajectory in recent years. Industrial engineering and electrical-electronics engineering have tracked each other closely throughout, sitting in the mid-range. Civil engineering consistently sits at the bottom of the group and shows the widest dip in 2021, suggesting it was more sensitive to whatever data issue affected that year. The overall trend across all sub-fields is a convergence upward in 2025, which may reflect a genuine increase in competition across engineering as a whole.

```
df |>
  filter(bolum_grubu %in% top_eng, year >= 2022) |>
  ggplot(aes(x = puan, fill = bolum_grubu)) +
  geom_histogram(bins = 25, show.legend = FALSE, color = "white") +
```



```
facet_wrap(~bolum_grubu, scales = "free_y") +
labs(title = "Score Distribution of Top Engineering Sub-fields (2022-2025)",
     x = "Minimum Admission Score", y = "Count") +
theme_minimal()
```



Figure 6: Score distribution panels for the top 6 engineering sub-fields (2022-2025).

As shown in Figure 6, the panel view highlights the differences more clearly. Computer engineering has a wider distribution shifted toward higher scores, while civil engineering programs are clustered at the lower end. This confirms that the demand gap between sub-fields is not just about averages — it extends across the entire score range.

```
df |>
  filter(kategori == "Muhendislik",
         unitur %in% c("DEVLET", "VAKIF"),
         year == 2024) |>
  ggplot(aes(x = unitur, y = puan, fill = unitur)) +
  geom_boxplot(show.legend = FALSE) +
  scale_fill_manual(values = c("DEVLET" = "#2E86AB", "VAKIF" = "#F39C12")) +
  labs(title = "State vs Private: Engineering Score Distribution (2024)",
       x = NULL, y = "Minimum Admission Score") +
  theme_minimal()
```

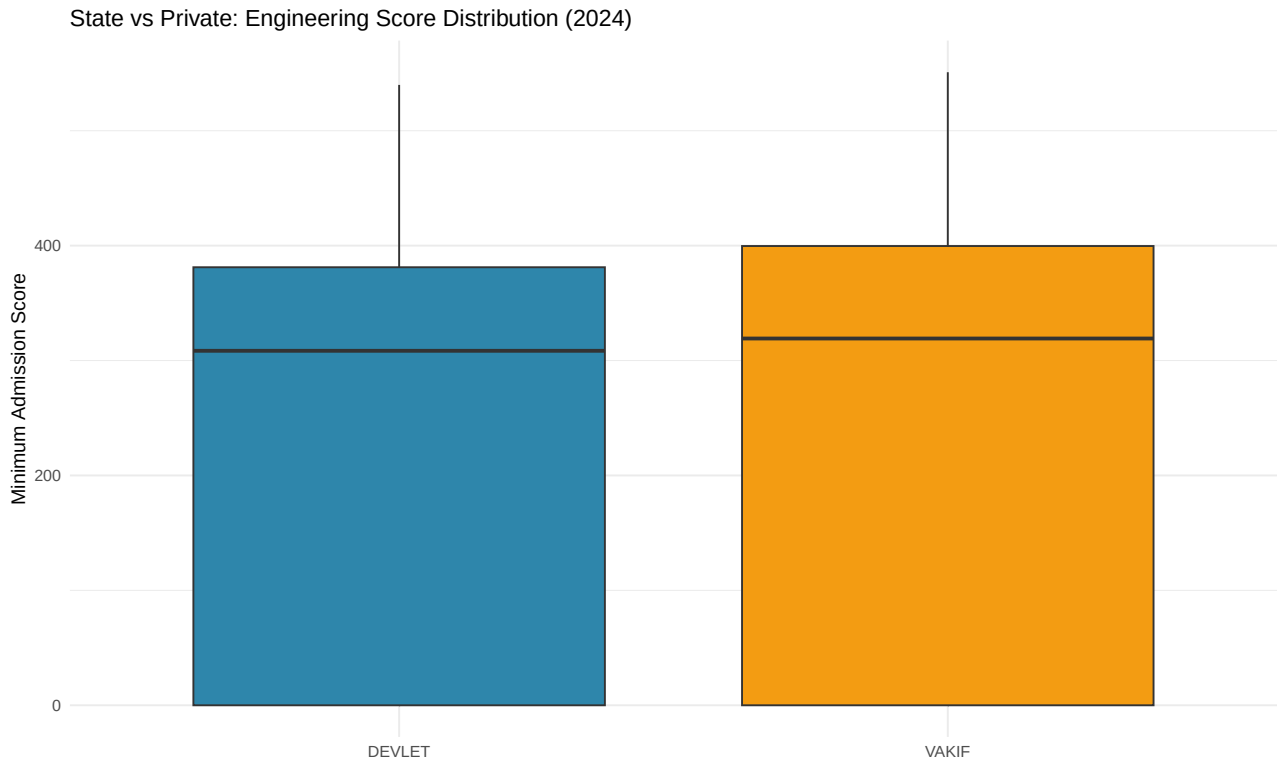


Figure 7: Minimum admission score distribution for state and private engineering programs in 2024.

As shown in Figure 7, contrary to what might be expected, the score distributions for state and private engineering programs in 2024 are quite similar in terms of median — both sit around 280–290. The main difference is in the spread: private university programs show a noticeably wider interquartile range and a longer lower tail, meaning there are more private programs with very low admission scores pulling their distribution down. State programs also include very low outliers, but their overall box is slightly more compact. The difference is real but not as dramatic as commonly assumed.

3.4 Regional Analysis

```
top_cities <- df |>
  filter(!is.na(il), il != "", year == 2024, kategori != "Diger") |>
  group_by(il) |>
  summarise(n_programs = n(),
            avg_puan = round(mean(puan, na.rm = TRUE), 1)) |>
  arrange(desc(n_programs)) |>
  head(10)

ggplot(top_cities, aes(x = reorder(il, n_programs), y = n_programs)) +
  geom_col(fill = "#2E86AB") +
  coord_flip() +
  labs(title = "Top 10 Cities by Number of Programs (2024)",
       x = NULL, y = "Number of Programs") +
  theme_minimal()
```

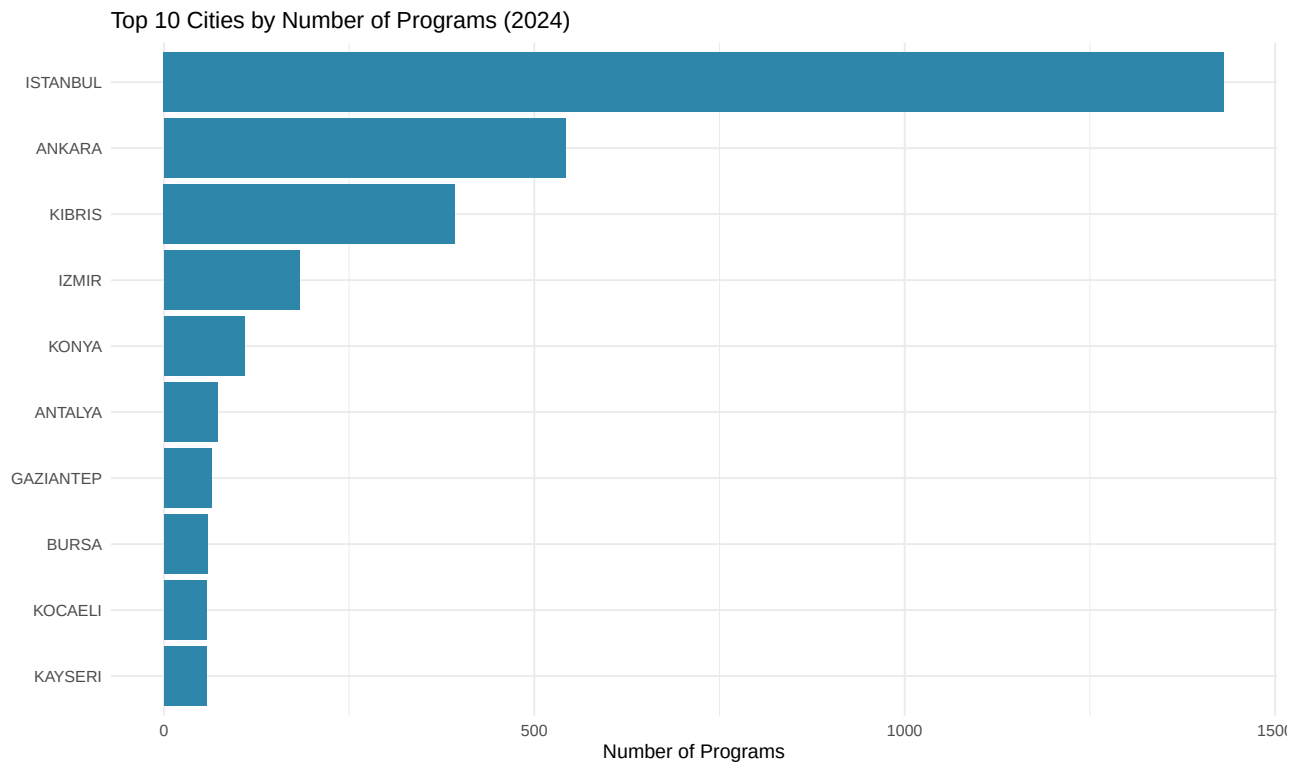


Figure 8: Top 10 cities in Turkey by number of university programs in 2024.

As shown in Figure 8, Istanbul leads by a very large margin with around 1,400 programs, more than double Ankara's roughly 580. What stands out unexpectedly is Kıbrıs (Northern Cyprus) in third place with about 390 programs, ahead of İzmir — this reflects the large number of Turkish-founded universities operating there that cater heavily to Turkish students. After Konya in fifth place, the remaining cities — Antalya, Gaziantep, Bursa, and Kayseri — are all clustered closely together with similar program counts, ranging roughly between 57 and 72.

3.5 Trend Prediction

```
# Simple linear model for Bilgisayar Muh.
bilgisayar <- df |>
  filter(bolum_grubu == "Bilgisayar Muh.") |>
  group_by(year) |>
  summarise(avg_puan = mean(puan, na.rm = TRUE))

model <- lm(avg_puan ~ year, data = bilgisayar)
cat("Linear model summary:\n")
```

Linear model summary:

```
cat(" Coefficient (yearly change):", round(coef(model)[2], 2), "\n")
```

Coefficient (yearly change): -3.38

```
cat(" R-squared:", round(summary(model)$r.squared, 3), "\n")
```

R-squared: 0.028

```
# Predict 2026
pred_2026 <- predict(model, newdata = data.frame(year = 2026))
cat(" Predicted 2026 avg score:", round(pred_2026, 1), "\n")
```

Predicted 2026 avg score: 332.6

```
ggplot(bilgisayar, aes(x = year, y = avg_puan)) +
  geom_point(size = 3, color = "#2E86AB") +
  geom_smooth(method = "lm", se = TRUE, color = "#E74C3C", linetype = "dashed") +
  scale_x_continuous(breaks = 2018:2026, limits = c(2018, 2026)) +
  labs(title = "Computer Engineering: Score Trend with Linear Prediction",
       x = "Year", y = "Average Min Score",
       subtitle = paste0("Predicted 2026 score: ", round(pred_2026, 1))) +
  theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'

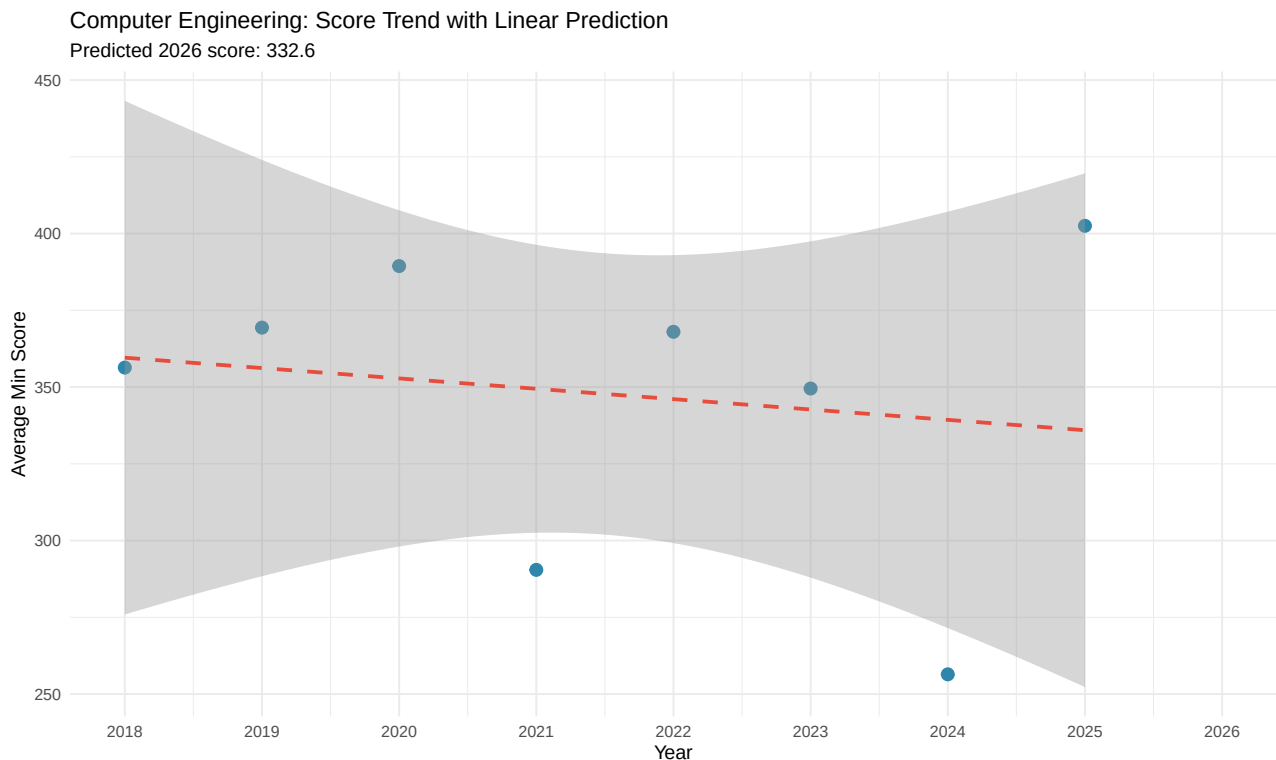


Figure 9: Linear trend model fitted to computer engineering average admission scores, with prediction for 2026.

As shown in Figure 9, the linear model fitted to computer engineering scores shows a slightly negative slope, meaning the trend line is gently declining over the 2018–2025 period. This is largely driven by the 2021 outlier and the low 2024 value, which pull the regression line down despite the strong

recovery in 2025. The predicted 2026 score of 332.6 is therefore lower than the most recent observed value of around 405, which suggests the model is not capturing the actual recent momentum well. The confidence interval is also very wide, indicating high uncertainty. This analysis serves as a useful exercise in trend modeling, but a more robust approach — such as excluding the anomalous 2021 data point or using a nonlinear model — would give a more reliable forecast.

4. Key Takeaways

- This study analyzed university admission trends in Turkey across seven field categories using YÖK Atlas data spanning 2018 to 2025, covering tens of thousands of program records from both state and private institutions.
- Medicine and dentistry/pharmacy consistently maintained the highest admission scores throughout the period, while engineering — despite being the largest category by program count — showed a wide internal spread driven by the inclusion of both highly selective and open-access programs.
- Computer engineering and software engineering emerged as the most competitive engineering sub-fields by 2025, while civil engineering remained at the lower end, reflecting a broader shift in student preferences toward technology-oriented careers.
- State and private university engineering programs showed more similar score distributions than expected, with the key difference being that private institutions include a longer tail of very low-scoring programs rather than a uniformly lower performance level.
- A simple linear model predicts a 2026 computer engineering score of 332.6, but the wide confidence interval and the influence of the anomalous 2021 data point make this estimate unreliable — the actual 2025 value of ~405 suggests the real trend may be stronger than the model captures.

References

- YÖK Atlas. Council of Higher Education, Turkey. <https://yokatlas.yok.gov.tr/>
- Cavus, M., & Aydin, O. *thestats: An R package for YÖK Atlas data*. <https://github.com/analyticsresearchlab/thestats>
- MorphaxTheDeveloper. *YÖK Atlas 2025 Dataset*. <https://github.com/MorphaxTheDeveloper/yokatlas-dataset-2025>

AI usage note: Claude was used to assist with data collection scripting, preprocessing, and code debugging. All analysis interpretations and written content belong to the authors.